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Avalanche Energy hits key milestone on the road to a desktop fusion reactor

For every technology, there comes a tipping point where it switches from theory to possibility. [Robin Langtry](#) believes that [Avalanche Energy](#) has reached that point.

Avalanche recently operated its desktop fusion machine for hours on end while maintaining 300,000 volts, a figure the startup predicts will allow it to build a reactor capable of generating more energy than it consumes, the holy grail for any [fusion company](#).

Where other fusion companies need powerful magnets to generate energy, Avalanche's design uses intense electrical currents to draw fast-moving ions into tight orbits around an electrode. As the density and speed of the ions rises, they begin to collide and fuse, releasing energy in the process.

"Getting to really high voltages is the key thing," Langtry, Avalanche's co-founder and CEO, told TechCrunch. And given that the company is building small reactors — targeting anywhere from 5 kilowatts to several hundred kilowatts — the density of that voltage, 6 million volts per meter, is important. "That's the real unlock for us," he said.

With that sort of force, Avalanche expects it will be able to generate a large number of neutrons at low cost, which can be used to make radioisotopes and to evaluate materials for use in fusion reactors.

IMAGE CREDITS: AVALANCHE ENERGY

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The company was recently awarded \$10 million from Washington State to build FusionWERX, a testing facility where other fusion companies and researchers can book time to study their own fusion technologies. Money for the grant comes from proceeds from the state's carbon marketplace.

"You can test your hardware, and then you get to leave with full ownership of your IP at the end of the day," Langtry said.

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Sales of radioisotopes and rentals of the FusionWERX facility should make Avalanche profitable in 2028, he said. Langtry is forecasting that the company will generate \$30 million to \$50 million in revenue in 2029.

With a path to revenue, and having hit the 300,000-volt milestone, Avalanche is rumored to be raising a Series B round. Langtry wouldn't comment specifically on the company's fundraising activities, though he did say that it's lining up the

money needed to fulfill the 50% cost-match requirement of the Washington State grant.

“We’ve already got a pretty good chunk of it lined up,” he said. Raising money to launch FusionWERX is “goal number one right now,” he added. “Then all the other stuff is going to fall in place as those pieces start getting built.”

Update 7/23 at 5:20 pm ET: Avalanche is rumored to be raising a Series B, not a Series A.

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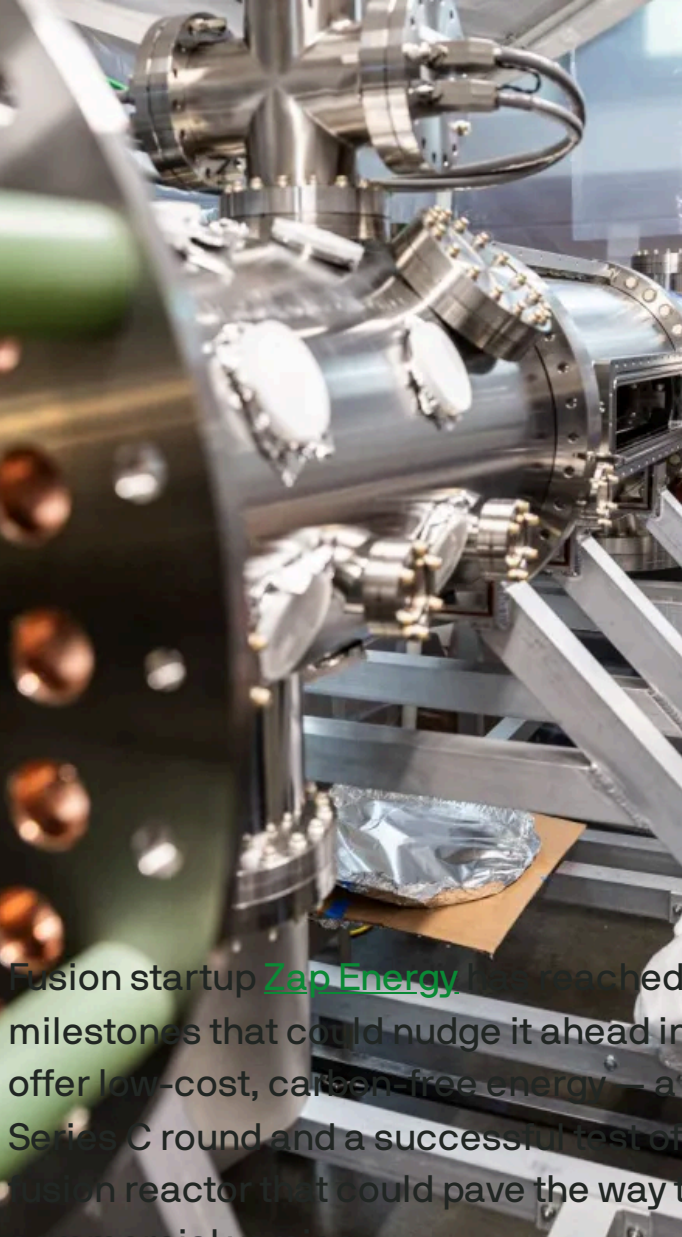
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Zap Energy nets \$160M Series C to advance its lightning-in-a-bottle fusion tech

Tim De Chant — 9:45 AM PDT · June 22, 2022

Fusion startup [Zap Energy](#) has reached two milestones that could nudge it ahead in the race to offer low-cost, carbon-free energy — a \$160 million Series C round and a successful test of a prototype fusion reactor that could pave the way to a commercial version.

IMAGE CREDITS: ZAP ENERGY

[Fusion power](#) has become an unlikely investor favorite as carbon emissions continue to rise and the effects of climate change become more apparent. We have been trying to harness the power of the sun to produce energy for many years, but after tens of billions of dollars and decades of research, fusion remains just out of reach.

Still, clever new approaches to contain scorching hot plasma — which burns at more than 100 million degrees Celsius — have brought fusion power tantalizingly close to reality. Investors are flocking to the field, hopeful that advances wrought by continued research and increasingly sophisticated computer simulations will finally help fusion pull away from its long string of failures.

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Zap Energy’s oversubscribed Series C was led by Lowercarbon Capital. New investors include Breakthrough Energy Ventures, Shell Ventures, DCVC and Valor Equity Partners. Existing investors Addition, Energy Impact Partners and Chevron Technology Ventures also contributed to the round. The startup’s core technology was spun out of research performed at the University of Washington and Lawrence Livermore National Laboratory.

Generally speaking, fusion power generates electricity by fusing hydrogen isotopes (either deuterium or tritium) into helium. The process releases neutrons, which are then captured to generate heat and spin a turbine. Atomic nuclei don’t like to fuse, so to coax them close enough for fusion to happen, nuclear scientists use extreme pressure and heat, creating a fourth state of matter known as plasma.

To create and contain the plasma, researchers generally take one of two approaches. In one, they compress hydrogen using [powerful magnetic fields](#) and then send an electric current through the resulting plasma to raise its temperature. In the other, they shoot a fuel pellet with a [powerful laser](#), which both compresses and heats the hydrogen isotopes.

Zap Energy takes a third approach known as a sheared flow stabilized Z-pinch. In the Z-pinch, engineers send a plasma stream through a vacuum chamber and then electrify it. The electric current flowing through the plasma creates a magnetic field that contains it (a tidy bit of recursion that

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ZAP ENERGY'S FUSION REACTOR LEVERAGES Z-PINCH PHYSICS, WHERE A LINE OF PLASMA CARRYING AN ELECTRICAL CURRENT GENERATES A MAGNETIC FIELD THAT CONFINES AND COMPRESSES — "PINCHES" — THE PLASMA.

IMAGE CREDITS: ZAP ENERGY

The electrified plasma in

Zap Energy's reactor design is strikingly similar to what happens in a thunderstorm. Lightning bolts superheat the air around them, creating plasma, which is then pinched by the electricity flowing through it. Essentially, Zap Energy is using lightning in a bottle to trigger a fusion reaction.

Researchers have been able to fuse atoms on countless occasions using all three approaches, but the challenge for commercial fusion power has always been designing a reactor that produces more power than is used to run it.

Lasers, magnets and electric fields all require enormous amounts of power. For example, [ITER](#), the massive experimental reactor being built in France, is expected to draw up to 620 MW at its peak — enough to power about 400,000 homes, if briefly.

By using the plasma to create its own magnetic confinement field, Zap's engineers don't have to build costly magnets or lasers, nor do they have to

power them, reducing the amount of energy needed to reach breakeven.

Confining plasma, no matter how you go about it, is tricky business, though. Plasma is notoriously unstable under magnetic confinement and that instability is part of what has bedeviled attempts to achieve stable and net-positive fusion power. Early fusion experiments attempted to harness a variation of the Z-pinch but failed because the plasma grew unstable. Those failures drove researchers to largely favor magnetic or laser confinement.

But Zap Energy thinks it has solved the instabilities in the Z-pinch approach — its investors are clearly hoping that’s the case and that Z-pinch reactors will be both cheaper to build and faster to market. The company expects its reactor’s power production will reach the breakeven point within a year.

Fusion power has long been derided as always “just a few years away.” That’s still the case, but if Zap Energy can achieve breakeven and convincingly show that it can push past that, fusion proponents might finally prove their critics wrong.

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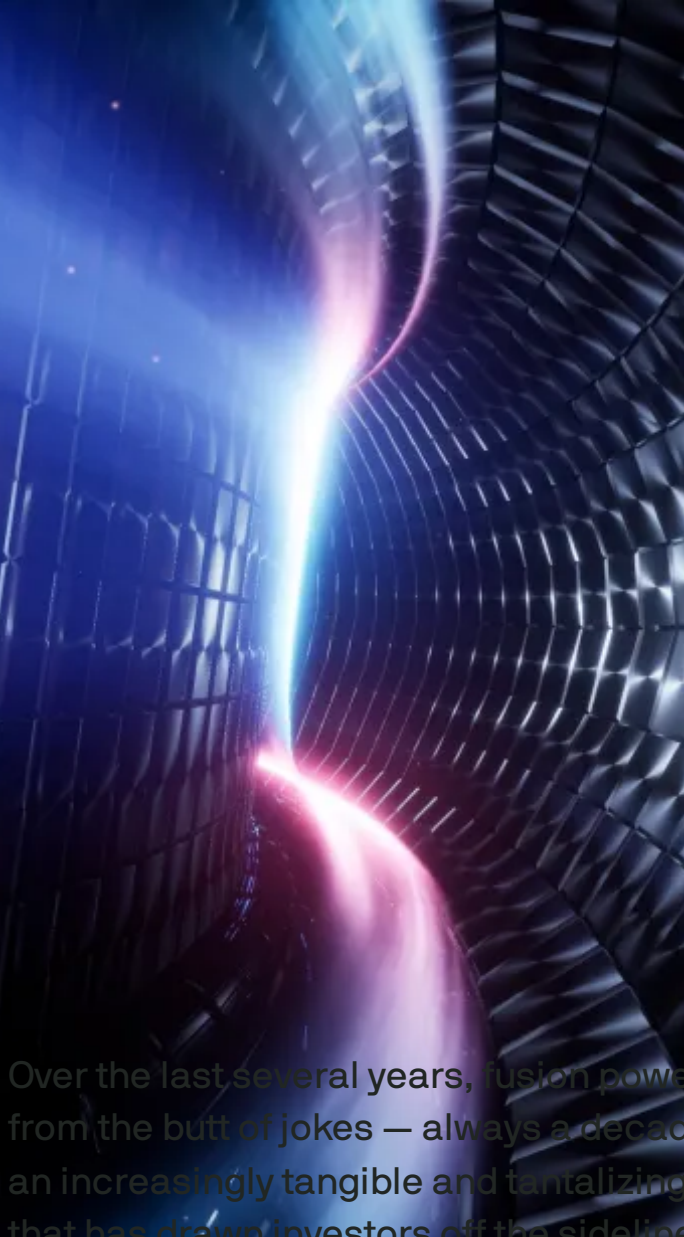
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Every fusion startup that has raised over \$100M

Tim De Chant — 8:39 AM PDT · June 19, 2025

Over the last several years, fusion power has gone from the butt of jokes — always a decade away! — to an increasingly tangible and tantalizing technology that has drawn investors off the sidelines.

IMAGE CREDITS: JOHN D / GETTY IMAGES

The technology may be challenging to master and expensive to build today, but fusion promises to harness the nuclear reaction that powers the sun to generate nearly limitless energy here on Earth. If startups are able to complete commercially viable fusion power plants, then they have the potential to upend trillion-dollar markets.

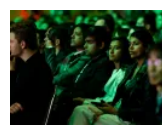
The bullish wave buoying the fusion industry has been driven by [three advances](#): more powerful computer chips, more sophisticated AI, and powerful high-temperature superconducting magnets. Together, they have helped deliver more sophisticated reactor designs, better simulations, and more complex control schemes.

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It doesn't hurt that, at the end of 2022, a U.S. Department of Energy lab announced that it had produced a controlled fusion reaction that produced [more power than the lasers had imparted](#) to the fuel pellet. The experiment had crossed what's known as scientific breakeven, and while it's still a long ways from commercial breakeven, where the reaction produces more than the entire facility consumes, it was a long-awaited step that proved the underlying science was sound.

Founders have built on that momentum in recent years, pushing the private fusion industry forward at a rapid pace.

Commonwealth Fusion Systems

With a \$1.8 billion Series B, [Commonwealth Fusion Systems](#) catapulted itself into the pole position in 2021. Since then, the company has been quiet on the fundraising front (no surprise), but it has been hard at work in Massachusetts building Sparc, its first-of-a-kind power plant intended to produce power at what it calls "commercially relevant" levels.

Sparc's reactor uses a tokamak design, which resembles a doughnut. The D-shaped cross section is wound with [high-temperature superconducting tape](#), which when energized, generates a powerful magnetic field that will contain and compress the superheated plasma. In Sparc's successor, the commercial-scale Arc, heat generated from the reaction is converted to steam to power a turbine. CFS designed its magnets in collaboration with MIT, where co-founder and CEO Bob Mumgaard worked as a researcher on fusion reactor designs and high-temperature superconductors.

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This Detroit startup is turning to utilities to make home efficiency upgrades cheaper

Tim De Chant - Aug 1, 2025



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Backed by Breakthrough Energy Ventures, The Engine, Bill Gates, and others, Devens, Massachusetts-based CFS expects to have Arc operational in the early 2030s. The company has raised a total of \$2 billion, according to PitchBook.

TAE

Founded in 1998, [TAE Technologies](#) (formerly known as Tri Alpha Energy) was spun out of the University of California, Irvine by Norman Rostoker. It uses a field-reversed configuration, but with a twist: after the two plasma shots collide in the middle of the reactor, the company bombards the plasma with particle beams to keep it spinning in a cigar shape. That improves the stability of the plasma, allowing more time for fusion to occur and for more heat to be extracted to spin a turbine.

The company raised [\\$150 million in June](#) from existing investors, including Google, Chevron, and

New Enterprise. TAE has raised \$1.79 billion in total, according to PitchBook.

Helion

Of all fusion startups, [Helion](#) has the most aggressive timeline. The company plans to produce electricity from its reactor in 2028. Its first customer? Microsoft.

Helion, based in Everett, Washington, uses a type of reactor called a field-reversed configuration, where magnets surround a reaction chamber that looks like an hourglass with a bulge at the point where the two sides come together. At each end of the hourglass, they spin the plasma into doughnut shapes that are shot toward each other at more than 1 million mph. When they collide in the middle, additional magnets help induce fusion. When fusion occurs, it boosts the plasma's own magnetic field, which induces an electrical current inside the reactor's magnetic coils. That electricity is then harvested directly from the machine.

The company [raised \\$425 million](#) in January 2025, around the same time that it turned on Polaris, a prototype reactor. Helion has raised \$1.03 billion, according to PitchBook. Investors include Sam Altman, Reid Hoffman, KKR, BlackRock, Peter Thiel's Mithril Capital Management, and Capricorn Investment Group.

Pacific Fusion

[Pacific Fusion](#) burst out of the gate with a \$900 million Series A, a whopping sum even among well-funded fusion startups. The company will use inertial confinement to achieve fusion, but instead of lasers compressing the fuel, it will use [coordinated electromagnetic pulses](#). The trick is in

the timing: All 156 impedance-matched Marx generators need to produce 2 terawatts for 100 nanoseconds, and those pulses need to simultaneously converge on the target.

The company is led by CEO Eric Lander, the scientist who led the Human Genome Project, and president Will Regan. Pacific Fusion's funding might be massive, but the startup hasn't gotten it all at once. Rather, its investors will pay out in tranches when the company achieves specified milestones, an approach that's common in biotech.

Shine Technologies

[Shine Technologies](#) is taking a cautious — and possibly pragmatic — approach to generating fusion power. Selling electrons from a fusion power plant is years off, so instead, it's starting by selling neutron testing and medical isotopes. More recently, it has been developing a way to recycle radioactive waste. Shine hasn't picked an approach for a future fusion reactor, instead saying that it's developing necessary skills for when that time comes.

The company has raised a total of \$778 million, according to PitchBook. Investors include Energy Ventures Group, Koch Disruptive Technologies, Nucleation Capital, and the Wisconsin Alumni Research Foundation.

General Fusion

Now its third-decade, [General Fusion](#) has raised \$440.53 million, according to PitchBook. The Richmond, British Columbia-based company was founded in 2002 by physicist Michel Laberge, who wanted to prove a different approach to fusion known as magnetized target fusion (MTF). Investors

include Jeff Bezos, Temasek, BDC Capital, and Chrysalix Venture Capital.

In an General Fusion's reactor, a liquid metal wall surrounds a chamber in which plasma is injected. Pistons surrounding the wall push it inward, compressing the plasma inside and sparking a fusion reaction. The resulting neutrons heat the liquid metal, which can be circulated through a heat exchanger to generate steam to spin a turbine.

General Fusion hit a rough patch in spring 2025. The company ran short of cash as it was building LM26, its latest device that it hoped would hit breakeven in 2026. Just days after hitting a key milestone, it [laid off 25% of its staff](#).

Tokamak Energy

[Tokamak Energy](#) takes the usual tokamak design — the doughnut shape — and squeezes it, reducing its aspect ratio to the point where the outer bounds start resembling a sphere. Like many other tokamak-based startups, the company uses high-temperature superconducting magnets (of the rare earth barium copper oxide, or REBCO, variety). Since its design is more compact than a traditional tokamak, it requires less in the way of magnets, which should reduce costs.

The Oxfordshire, U.K.-based startup's ST40 prototype, which looks like a large, steampunk Fabergé egg, generated an ultra-hot, 100 million degree C plasma in 2022. Its next generation, Demo 4, is currently under construction and is intended to test the company's magnets in "fusion power plant-relevant scenarios." Tokamak Energy raised [\\$125 million in November 2024](#) to continue its reactor design efforts and expand its magnet business.

In total, the company has raised \$336 million from investors including Future Planet Capital, In-Q-Tel, Midven, and Capri-Sun founder Hans-Peter Wild, according to PitchBook.

Zap Energy

[Zap Energy](#) isn't using high-temperature superconducting magnets or super-powerful lasers to keep its plasma confined. Rather, it zaps the plasma (get it?) with an electric current, which then generates its own magnetic field. The magnetic field compresses the plasma about 1 millimeter, at which point ignition occurs. The neutrons released by the fusion reaction bombard a liquid metal blanket that surrounds the reactor, heating it up. The liquid metal is then cycled through a heat exchanger, where it produces steam to drive a turbine.

Like Helion, Zap Energy is based in Everett, Washington, and the company has raised \$327 million, according to PitchBook. Backers include Bill Gates' Breakthrough Energy Ventures, DCVC, Lowercarbon, Energy Impact Partners, Chevron Technology Ventures, and Bill Gates as an angel.

Proxima Fusion

Most investors have favored large startups that are pursuing tokamak designs or some flavor of inertial confinement. But stellarators have shown great promise in scientific experiments, including the Wendelstein 7-X reactor in Germany.

[Proxima Fusion](#) is bucking the trend, though, having attracted a [€130 million Series A](#) that brings its total raised to more than €185 million. Investors include Balderton Capital and Cherry Ventures.

Stellarators are similar to tokamaks in that they confine plasma in a ring-like shape using powerful magnets. But they do it with a twist — literally. Rather than force plasma into a human-designed ring, stellarators twist and bulge to accommodate the plasma's quirks. The result should be a plasma that remains stable for longer, increasing the chances of fusion reactions.

Marvel Fusion

[Marvel Fusion](#) follows the inertial confinement approach, the same basic technique that the National Ignition Facility used to prove that controlled nuclear fusion reactions could produce more power than was needed to kick them off. Marvel [fires powerful lasers](#) at a target embedded with silicon nanostructures that cascade under the bombardment, compressing the fuel to the point of ignition. Because the target is made using silicon, it should be relatively simple to manufacture, leaning on the semiconductor manufacturing industry's decades of experience.

The inertial confinement fusion startup is building a demonstration facility in collaboration with Colorado State University, which it expects to have operational by 2027. Munich-based Marvel has raised a total of \$161 million from investors including b2venture, Deutsche Telekom, Earlybird, HV Capital, and Taavet Hinrikus and Albert Wenger as angels.

First Light

[First Light](#) dropped its pursuit of fusion power in March 2025, pivoting instead to become a technology supplier to fusion startups and other companies. The startup had previously followed an

approach known as inertial confinement, in which fusion fuel pellets are compressed until they ignite.

First Light, which is based in Oxfordshire, U.K., has raised \$140 million, according to PitchBook, from investors including Invesco, IP Group, and Tencent.

Xcimer

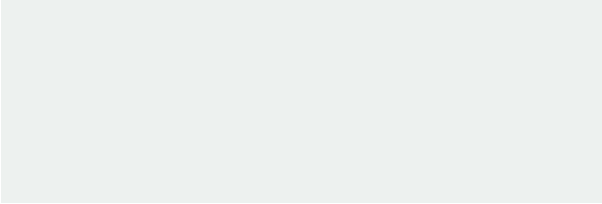
Though nothing about fusion can be described as simple, [Xcimer](#) takes a relatively straightforward approach: follow the basic science that's behind the National Ignition Facility's breakthrough net-positive experiment, and [redesign the technology that underpins it](#) from the ground up. The Colorado-based startup is aiming for a 10-megajoule laser system, five times more powerful than NIF's setup that made history. Molten salt walls surround the reaction chamber, absorbing heat and protecting the first solid wall from damage.

Founded in January 2022, Xcimer has already raised \$109 million, according to PitchBook, from investors including Hedosophia, Breakthrough Energy Ventures, Emerson Collective, Gigascale Capital, and Lowercarbon Capital.

This story was originally published in September 2024 and will be continually updated.

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